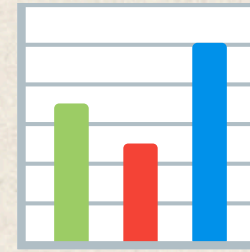


PIZZA



 **PIZZA SALES ANALYSIS**
SQL DATA ANALYSIS PROJECT
 **PREPARED BY**
ROHIT SHAW
 **ASPIRING DATA ANALYST**



SQL Pizza Sales Analysis Project

**A Data Analytics
SQL Project**

About Our Pizza Place

48620

Total_quantity

4.9

Good Rate

+100

Pizza menu

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Tools Used

- **MySQL**
- **MySQL Workbench**
- **Excel**



Project Overview Objective

The objective of this project is to analyze pizza sales data using SQL and extract meaningful business insights that can help improve sales performance and decision-making

Database : Pizza Sales Database



Dataset :

- **Orders Table**
- **Order Details Table**
- **Pizza Table**
- **Pizza Types Table**

Total Tables : 4

Database Schema

Orders

order_id
date
time

Order_details

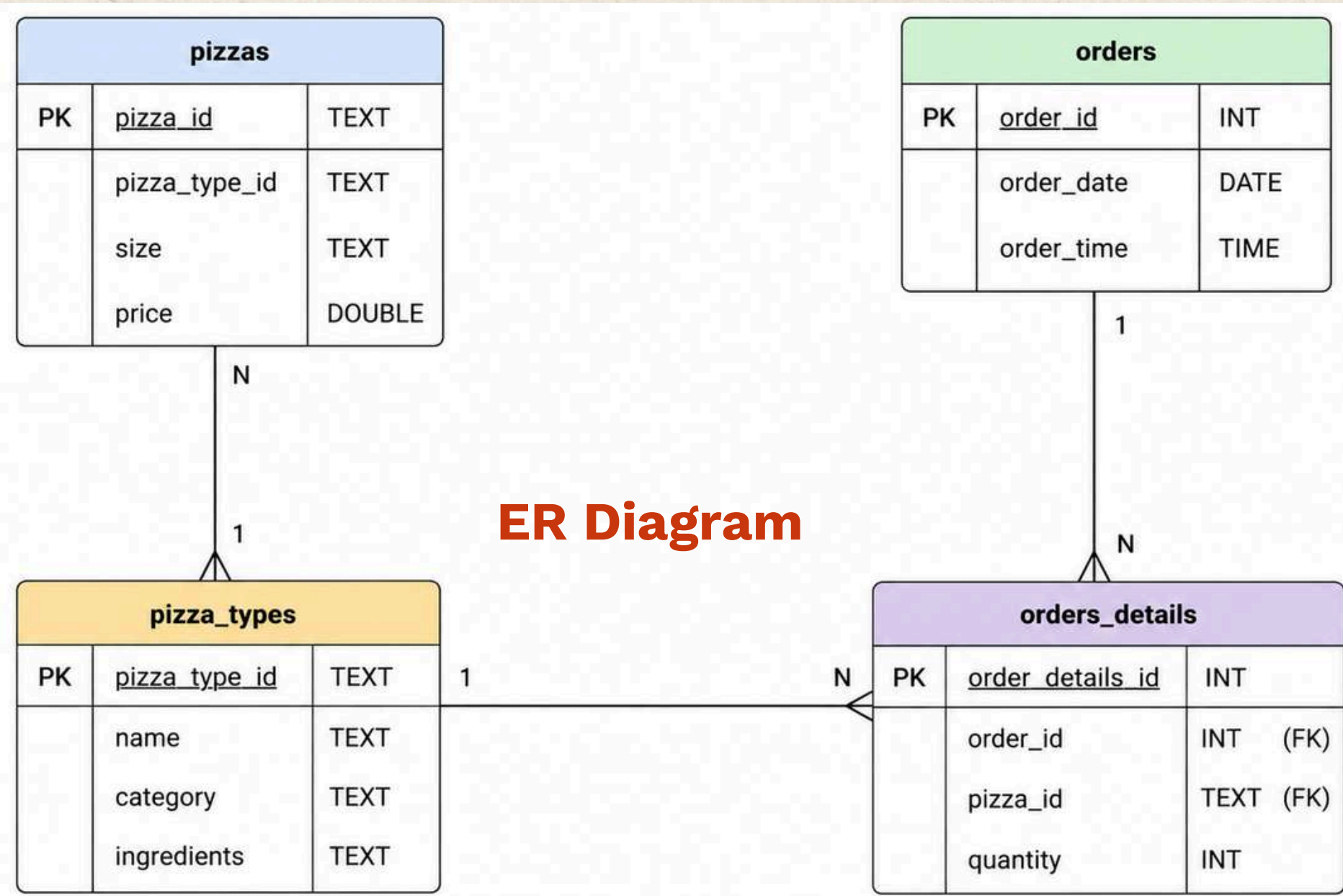
order_details_id
order_id
pizza_id
quantity

Pizzas

pizza_id
pizza_type_id
size
price

Pizza_types

pizza_type_id
name
category
ingredients



Real scenario Based Business Questions

QUESTION LEVEL 1

- 1 RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.
- 2 CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.
- 3 IDENTIFY THE HIGHEST-PRICED PIZZA.
- 4 IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.
- 5 LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

INTERMEDIATE: - LEVEL -2

- 6 . JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.
- 7 . DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.
- 8 . JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.
- 9 . GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.
- 10 . DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

ADVANCED: LEVEL - 3

- 11 . CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.
12. . ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.
- 13 . DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.



1 .Retrieve the total number of orders placed?



```
SELECT  
    COUNT(order_id) AS Totsl_Orders  
FROM  
    orders;
```

Result Grid			
	Totsl_Orders		
▶	21350		

INSIGHT ; TOTAL ORDERS = 21350

2 Calculate the total revenue generated from pizza sales.

SELECT ALL

```
ROUND(SUM(orders_details.quantity * pizzas.price),  
2) AS Total_Revenue
```

FROM

```
orders_details
```

INNER JOIN

```
pizzas ON orders_details.pizza_id = pizzas.pizza_id;
```

Result Grid

	Total_Revenue
▶	817860.05



3. Identify the highest-priced pizza.



```
SELECT
    pizza_types.name, MAX(price) AS Highest_Price
FROM
    pizza_types
    INNER JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.name
ORDER BY Highest_Price DESC
LIMIT 1;
```

**The Greek Pizza has the Highest =
35.39**

Result Grid			Filter Rows:
	name	Highest_Price	
▶	The Greek Pizza	35.95	

4 . Identify the most common pizza size ordered.

```
SELECT DISTINCT
    pizzas.size,
    ROUND(COUNT(orders_details.quantity), 0) AS Total_size_BY_Orders
FROM
    orders_details
    INNER JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizzas.size;
```

Result Grid		Filter Rows:
	size	Total_size_BY_Orders
▶	M	15385
	L	18526
	S	14137
	XL	544
	XXL	28



5. List the top 5 most ordered pizza types along with their quantities.?

```
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS Top_5_Order_Pizza
FROM
    orders_details
    JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
    JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.name
ORDER BY SUM(orders_details.quantity) DESC
LIMIT 5;
```

	name	Top_5_Order_Pizza
	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

6) Join the necessary tables to find the total quantity of each pizza category ordered.?



```
SELECT
    pizza_types.category, SUM(orders_details.quantity)
FROM
    orders_details
    INNER JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
    INNER JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.category
ORDER BY SUM(orders_details.quantity) DESC;
```

INSIGHT : ALL CATEGORY

	category	SUM(orders_details.quantity)
►	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

7. Determine the distribution of orders by hour of the day.?

```
SELECT
    HOUR(order_time) AS Hours,
    COUNT(orders.order_id) AS orders_count
FROM
    orders
    INNER JOIN
    orders_details ON orders.order_id = orders_details.order_id
GROUP BY HOUR(order_time)
ORDER BY COUNT(orders.order_id) ASC;
```

INSIGHT : ORDERS BY HOUR OF THE DAY



	Hours	orders_count
▶	12	6543
	13	6203
	18	5359
	17	5143
	19	4350
	16	4185
	14	3521
	20	3487
	15	3170

8. Join relevant tables to find the category-wise distribution of pizzas.?

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

	category	COUNT(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9



9. Group the orders by date and calculate the average number of pizzas ordered per day.?

```
SELECT
```

```
    orders.order_date, SUM(orders_details.quantity)
```

```
FROM
```

```
    orders
```

```
    JOIN
```

```
    orders_details ON orders.order_id = orders_details.order_id
```

```
GROUP BY orders.order_date;
```



	order_date	SUM(orders_details.quantity)
▶	2015-01-01	162
	2015-01-02	165
	2015-01-03	158
	2015-01-04	106
	2015-01-05	125
	2015-01-06	147
	2015-01-07	138

Result 15 ×

10. Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    AVG(quantity)
FROM
    (SELECT
        orders.order_date, SUM(orders_details.quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid	
	avg(quantity)
▶	138.4749



11. Determine the top 3 most ordered pizza types based on revenue.?



```
SELECT
  pizza_types.name,
  SUM(orders_details.quantity * pizzas.price) AS Top_3_Revenue
FROM
  pizza_types
  JOIN
  pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
  JOIN
  orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY SUM(orders_details.quantity * pizzas.price) DESC
LIMIT 3;
```

Result Grid			Filter Rows:
	name	Top_3_Revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	

12) Calculate the percentage contribution of each pizza type to total revenue.?



```
SELECT
    pizza_types.category,
    ROUND(SUM(orders_details.quantity * pizzas.price), 2) AS Total_Revenue,
    ROUND(SUM(orders_details.quantity * pizzas.price) *100/sum(SUM(orders_details.quantity * pizzas.price)) over (),2) AS Total_Percentage
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY Total_Revenue DESC;
```

Result Grid			
Filter Rows:			
	category	Total_Revenue	Total_Percentage
▶	Classic	220053.1	26.91
	Supreme	208197	25.46
	Chicken	195919.5	23.96
	Veggie	193690.45	23.68



13. Analyze the cumulative revenue generated over time.?

```
SELECT
  orders.order_date,
  ROUND(SUM(orders_details.quantity * pizzas.price), 2) AS revenue,
  ROUND(sum(SUM(orders_details.quantity * pizzas.price))
  OVER (ORDER BY orders.order_date), 2) AS Cumulative_Total
FROM
  orders_details
  JOIN
  pizzas ON orders_details.pizza_id = pizzas.pizza_id
  JOIN
  orders ON orders.order_id = orders_details.order_id
GROUP BY orders.order_date
ORDER BY Cumulative_Total asc ;
```

Result Grid				Filter Rows:	
	order_date	revenue	Cumulative_Total		
▶	2015-01-01	2713.85	2713.85		
	2015-01-02	2731.9	5445.75		
	2015-01-03	2662.4	8108.15		
	2015-01-04	1755.45	9863.6		
	2015-01-05	2065.95	11929.55		
	2015-01-06	2428.95	14358.5		
	2015-01-07	2202.2	16560.7		
	2015-01-08	2838.35	19399.05		
	2015-01-09	2127.35	21526.4		
	2015-01-10	2463.95	23990.35		
	2015-01-11	1872.3	25862.65		
	2015-01-12	1919.05	27781.7	27781	
	2015-01-13	2049.6	29831.3		
	2015-01-14	2527.4	32358.7		
	2015-01-15	1984.8	34343.5		



14. Determine the top 3 most ordered pizza types based on revenue for each pizza category ?

SELECT

```
    pizza_types.category,  
    pizza_types.name,  
    SUM(orders_details.quantity * pizzas.price) AS revenue
```

FROM

```
    pizza_types
```

JOIN

```
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
```

JOIN

```
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
```

GROUP BY pizza_types.category , pizza_types.name

order by revenue **desc limit** 3;

Result Grid				Filter Rows:	Exp
	category	name	revenue		
▶	Chicken	The Thai Chicken Pizza	43434.25		
	Chicken	The Barbecue Chicken Pizza	42768		
	Chicken	The California Chicken Pizza	41409.5		

Conclusion

This project demonstrates how SQL can be used to analyze sales data and extract meaningful business insights. By using SQL concepts such as JOINS, GROUP BY, Aggregate Functions, ORDER BY, and Window Functions, I analyzed pizza sales to identify customer preferences, revenue trends, and best-selling products. The insights generated from this analysis can help businesses make better decisions related to inventory, pricing, and sales strategies



- ✓ • **Successfully analyzed pizza sales data using SQL.**
- ✓ • **Solved real-world business questions using SQL queries.**
- ✓ • **Identified key sales trends and customer preferences.**
- ✓ • **Improved practical knowledge of SQL for data analysis.**
- ✓ • **Demonstrated how SQL supports data-driven business decisions**



Thank You !

Rohit Sahw



Data Analyst

Skills Demonstrated

- **SQL**
- **MySQL**
- **Data Analysis**

Business Insights

End Slide